AN ETALE MORPHISM OF ALGEBRAIC SPACES WHOSE FIBRES CONTAIN NONTRIVIAL SPECIALIZATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The specialization chain lives in |U| (the etale cover), which is exactly the shape the counted statement asks for: specializations inside a fibre of an etale morphism of algebraic spaces.

Editorial note. The label lemma-specializations-fibre-etale names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 3993-4008: P = k[u, u^{-1}, y, x_n for n in Z] with the Z-action generated by tau(u) = u, tau(y) = u y, tau(x_n) = x_{n+1}; V(I_d) with I_d = ((1 - u^d) y, x_n - x_{n+d}) is the fix locus of tau^d; localizing at S generated by y and all 1 - u^d makes Z act freely on U = Spec(S^{-1}P); X = U/Z is the quotient algebraic space (Spaces, definition-quotient)
- Context span 4010-4018: the primes p_n = (x_n, x_{n+1}, ...) in S^{-1}P satisfy tau(p_n) = p_{n+1}, so they define points xi_n of U with one common image x in X; the specializations xi_n leads to xi_{n-1} hold in U; the context concludes U -> X is an example of the promised type

Source reference. Original definition 02Z3: sources/stacks-chow/tags.txt, source lines 3898–3898. Prerequisite pointer invoked at 4045; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

To give an example, let k be a field. Set

$$P = k[u, u^{-1}, y, \{x_n\}_{n \in \mathbf{Z}}].$$

Consider the action of $\mathbf{Z}$ on P by k-algebra maps generated by the automorphism τ given by the rules $\tau(u) = u$, $\tau(y) = uy$, and $\tau(x_n) = x_{n+1}$. For $d \geq 1$ set $I_d = ((1 - u^d)y, x_n - x_{n+d}, n \in \mathbf{Z})$. Then $V(I_d) \subset \mathrm{Spec}(P)$ is the fix point locus of τ^d. Let $S \subset P$ be the multiplicative subset generated by y and all $1 - u^d$, $d \in \mathbf{N}$. Then we see that $\mathbf{Z}$ acts freely on $U = \mathrm{Spec}(S^{-1}P)$. Let $X = U/\mathbf{Z}$ be the quotient algebraic space, see Spaces, Definition 02Z3.

Consider the prime ideals $\mathfrak{p}_n = (x_n, x_{n+1}, \ldots)$ in $S^{-1}P$. Note that $\tau(\mathfrak{p}_n) = \mathfrak{p}_{n+1}$. Hence each of these define point $\xi_n \in U$ whose image in X is the same point x of

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X. Moreover we have the specializations

$$\ldots \rightsquigarrow \xi_n \rightsquigarrow \xi_{n-1} \rightsquigarrow \ldots$$

We conclude that $U \to X$ is an example of the promised type.

Lemma 0.1. *There exists an étale morphism of algebraic spaces $f : X \to Y$ and a nontrivial specialization of points $x \rightsquigarrow x'$ in $|X|$ with $f(x) = f(x')$ in $|Y|$.*

Proof. See discussion above. □